Math 271: Mathematical Methods
Semester I, 2026-27

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(a) [24 marks]

Let $f : \mathbb{R}^n \to \mathbb{R}$, and, for any $x, y \in \mathbb{R}^n$, consider the function $\phi : [0,1] \to \mathbb{R}$ defined by $\phi(t) = f(tx + (1-t)y)$. Show that if $f : \mathbb{R}^n \to \mathbb{R}$ is *concave*, then $\phi : [0,1] \to \mathbb{R}$ is also *concave* for each $x, y \in \mathbb{R}^n$.

(b) [24 marks]

Let g be a C^1 function on an open interval I in $\mathbb{R}$. Prove that if g is *concave* on I then $g(y) - g(x) \leq g'(x) \cdot (y - x)$, for all $x, y \in I$.

(c) [24 marks]

Let F be a C^1 function on an open convex subset A of $\mathbb{R}^n$. Prove that if F is *concave* on A then $F(y) - F(x) \leq \nabla F(x) \cdot (y - x)$, for all $x, y \in A$.

(d) [28 marks]

Let F be a C^2 function on an open convex subset A of $\mathbb{R}^n$. Prove that if F is *concave* on A then the Hessian matrix $H_f(x)$ is *negative semidefinite* for all $x \in A$.